



TECH NEWS

TECHNOLOGY UPDATES

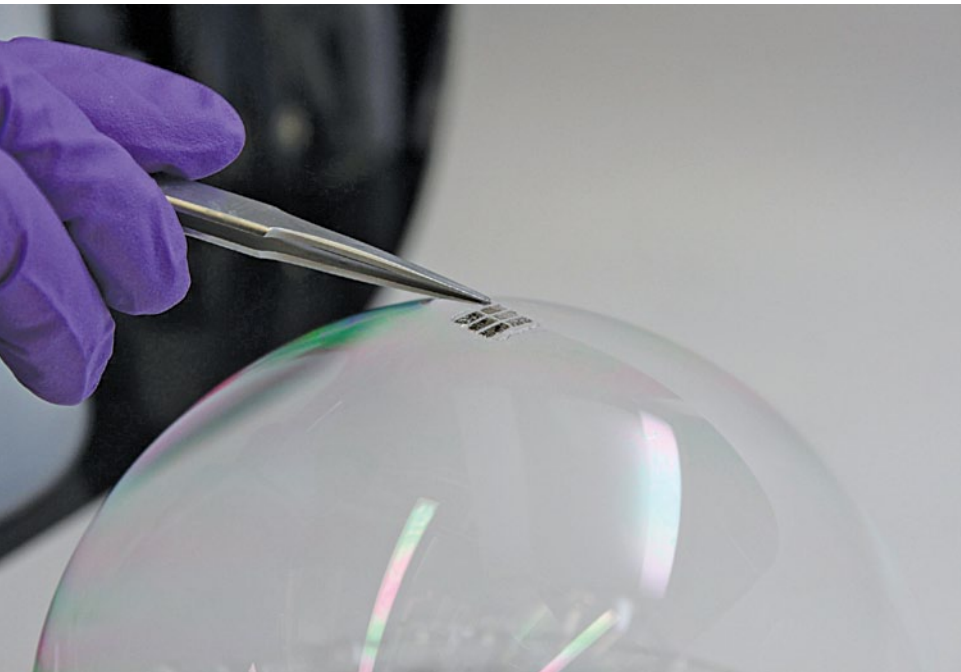
Solar cells developed as thin as a soap bubble

Engineers at Massachusetts Institute of Technology, USA, have developed the thinnest-ever solar cell, which is as

light as a soap bubble. This achievement might completely revolutionise the contemporary trend of miniaturising technology. The team prepared the chip by using a flexible polymer named parylene for both the substrate and the protective over-coating and an organic material called DBP as the light absorbent.

An author of the research paper explains that the goal was to produce a complete solar cell system with an artificial shield to screen the effects of the surroundings. Key approach of the study was to develop both the substrate and the device in the same system.

To achieve this, the researchers helped both the substrate and the solar cell to grow inside a vacuum tube where the temperature was set at room temperature and the effect of external solvents was negligible. The whole process was built upon vapour deposition technique.

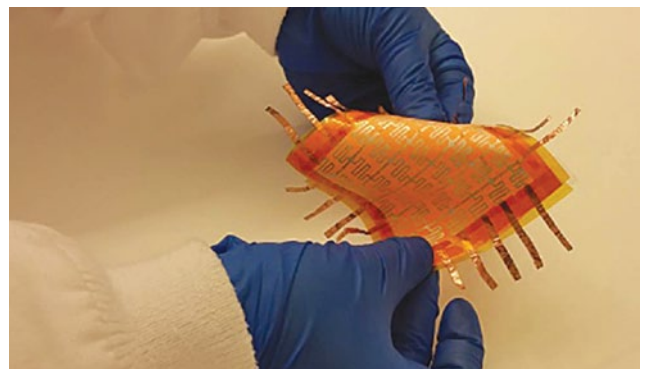


The ultra-thin solar cell system

Smart paper skin can imitate the functions of human skin

A group of electrical engineers from King Abdullah University of Science and Technology in Saudi Arabia has designed a sensor circuitry that can imitate the functions of human skin. It is flexible, paper based artificial skin, made using items found in a typical household, that is capable of sensing pressure, temperature, humidity, proximity, pH and air flow.

The multi-sensor artificial skin can respond to external stimuli and could have important applications in medicine and robotics. The paper based skin is layered onto a Post-it note, with paper, aluminium foil, lint-free wipes and pencil lines acting as sensing components. It is made of recyclable materials and presents a large number of sensory functions in a cheap and environmentally-friendly way.



The flexible temperature array was made by drawing a resistor structure with a silver conductive ink pen on Post-it paper (Image courtesy: King Abdullah University of Science and Technology)